along the rising portion of each edge. Notice that both the Bessel and the Chebyshev filter have removed the burst of high frequency noise present in the original signal.

The last choice is to use no antialias filter at all, as is shown in (d). This has the strong advantage that the value of each sample is *identical* to the value of the original analog signal. In other words, it has perfect edge sharpness; a change in the original signal is immediately mirrored in the digital data. The disadvantage is that aliasing can distort the signal. This takes two different forms. First, high frequency interference and noise, such as the example sinusoidal burst, will turn into meaningless samples, as shown in (d). That is, any high frequency noise present in the analog signal will appear as aliased noise in the digital signal. In a more general sense, this is not a problem of the sampling, but a problem of the upstream analog electronics. It is not the ADC's purpose to reduce noise and interference; this is the responsibility of the analog electronics before the digitization takes place. It may turn out that a Bessel filter should be placed before the digitizer to control this problem. However, this means the filter should be viewed as part of the analog processing, not something that is being done for the sake of the digitizer.

The second manifestation of aliasing is more subtle. When an event occurs in the analog signal (such as an edge), the digital signal in (d) detects the change on the *next* sample. There is no information in the digital data to indicate what happens *between* samples. Now, compare using *no filter* with using a *Bessel filter* for this problem. For example, imagine drawing straight lines between the samples in (c). The time when this constructed line crosses one-half the amplitude of the step provides a *subsample* estimate of when the edge occurred in the analog signal. When no filter is used, this subsample information is completely lost. You don't need a fancy theorem to evaluate how this will affect your particular situation, just a good understanding of what you plan to do with the data once is it acquired.

Multirate Data Conversion

There is a strong trend in electronics to replace *analog circuitry* with *digital algorithms*. Data conversion is an excellent example of this. Consider the design of a digital voice recorder, a system that will digitize a voice signal, store the data in digital form, and later reconstruct the signal for playback. To recreate intelligible speech, the system must capture the frequencies between about 100 and 3000 hertz. However, the analog signal produced by the microphone also contains much higher frequencies, say to 40 kHz. The brute force approach is to pass the analog signal through an eight pole low-pass Chebyshev filter at 3 kHz, and then sample at 8 kHz. On the other end, the DAC reconstructs the analog signal at 8 kHz with a zeroth order hold. Another Chebyshev filter at 3 kHz is used to produce the final voice signal.